

Full affirmative answer to the subcritical affine question

arXiv:2210.14798, Remark 20

Status. literature_implied_answer (full, in every well-defined finite-value case). A 2025 Dirichlet-range norm identity, combined with hyperbolic level-set monotonicity, supplies the ingredient absent from the 2022 proof. The supporting paper does not cite the source question or state this coherent-state consequence, so the identification and final hinge-majorization argument are recorded here explicitly.

Source question. For $\beta > 0$ put

$$\psi_{a,b}(x) = 2^\beta \Gamma(2\beta)^{-1/2} a^\beta x^{\beta-1/2} e^{-ax+ibx}, \quad a > 0, b \in \mathbb{R}.$$

Frank's Theorem 8 proves, when $\beta > 1/2$, that normalized coherent states maximize every convex functional of $|\langle \psi_{a,b}, \psi \rangle|^2$. Remark 20 asks whether this extends to $0 < \beta < 1/2$:

Also, the coherent states $\psi_{a,b}$, $(a,b) \in \mathbb{R}_+ \times \mathbb{R}$, are in one-to-one correspondence with the functions F_w , $w \in \mathbb{D}$. Indeed, a straightforward computation shows that the f corresponding to $\psi = \psi_{1,0}$ is $f(z) = (2i/(z+i))^{2\beta}$, which corresponds to $g(\zeta) = 1 = F_0(\zeta)$. The result in the general case follows from the facts that every (a,b) can be moved to $(1,0)$ by an $aX + b$ -action, every point $w \in \mathbb{D}$ can be moved to 0 by the action of a subgroup of $SU(1,1)$ isomorphic to $aX + b$ (see, e.g., [25, equation after (1.3)]) and that Σ relates these actions to each other.

Thus, Theorem 8 follows immediately from Theorem 10 in Case 3 with $\alpha = 2\beta$ and, similarly Corollary 9 follows from Corollary 14.

Remark 20. The functions $\psi_{a,b}$ are also well defined for $\beta \in (0, \frac{1}{2}]$. In this case, the coherent state transform cannot be normalized to be an isometry to a subset of $L^2(\mathbb{R}_+ \times \mathbb{R}, a^{-2} da db)$, but the optimization problem in Theorem 8 still makes sense. We claim that, for $\beta = \frac{1}{2}$, Theorem 8 remains valid, replacing the assumptions 'convex' and 'not linear' on Φ by 'nondecreasing' and 'strictly increasing near 1', respectively. Indeed, in this case the function f in the previous proof belongs to the Hardy space $H^2(\mathbb{C}_+)$ and, by Plancherel, its norm in that space is equal to $\|\psi\|_{L^2(\mathbb{R}_+)}$. Mapping $\mathbb{C}_+$ to $\mathbb{D}$ via Σ , we can deduce the assertion from Proposition 15. We do not know whether Theorem 8 extends to $\beta \in (0, \frac{1}{2})$.

4.6. Limitations of the method. In this paper we have discussed the cases of the Heisenberg group, $SU(2)$, $SU(1,1)$ and the affine group. It is a natural question, potentially of relevance for representation theory, to which extent the results can be

Finite-value convention. For $0 < \beta < 1/2$, the affine transform is not square-integrable, so linear functionals diverge. The nontrivial formulation is therefore the one in which the displayed coherent-state integral below is a finite real number. For a finite-valued convex Φ , this condition automatically forces $\Phi(0) = \Phi'_+(0) = 0$ and hence $\Phi \geq 0$. If instead $\Phi \geq 0$ and the coherent integral is $+\infty$, the supremum is also $+\infty$, already on a coherent state. Signed divergent integrals are not variational quantities and are not assigned a value here.

Theorem (Subcritical affine coherent-state majorization). *Let $0 < \beta < 1/2$ and let $\Phi : [0, 1] \rightarrow \mathbb{R}$ be convex. If*

$$\frac{2\pi}{\beta} \int_0^1 \Phi(s) s^{-1/(2\beta)-1} ds$$

is finite, then

$$\begin{aligned} \sup_{\substack{\psi \in L^2(\mathbb{R}_+) \\ \|\psi\|_2=1}} \iint_{\mathbb{R}_+ \times \mathbb{R}} \Phi(|\langle \psi_{a,b}, \psi \rangle|^2) \frac{da db}{a^2} \\ = \frac{2\pi}{\beta} \int_0^1 \Phi(s) s^{-1/(2\beta)-1} ds. \end{aligned}$$

Every normalized coherent state, up to phase, attains the supremum. If Φ is not affine linear, these are the only maximizers. The same equality holds with value $+\infty$ when $\Phi \geq 0$ and the right side diverges.

Proof intuition. Set $\alpha = 2\beta \in (0, 1)$. The disk model is no longer an ordinary weighted Bergman L^2 space; it is the Dirichlet-range RKHS with kernel $(1 - z\bar{w})^{-\alpha}$. This is exactly why the 2022 area-norm proof stops. Brevig–Kulikov–Seip–Zlotnikov later found the correct replacement: the RKHS norm is a renormalized weighted variation of the distribution of $|f|^{2/\alpha}(1 - |z|^2)$. Hyperbolic isoperimetry says the associated quantity $t(\mu(t) + 1)$ decreases. Once its total weighted variation is fixed, a one-variable calculation shows that every hinge $(u^2 - s)_+$ is largest for the constant disk function, equivalently a coherent state. Convex functions are positive mixtures of hinges.

Step 1: the exact disk Hilbert space. Let

$$c_\alpha(n) = \frac{\Gamma(n + \alpha)}{\Gamma(\alpha)\Gamma(n + 1)}, \quad \|f\|_{\alpha,2}^2 = \sum_{n=0}^{\infty} \frac{|\hat{f}(n)|^2}{c_\alpha(n)}.$$

This is the space A_α^2 of [2]; its reproducing kernel is $K_w(z) = (1 - z\bar{w})^{-\alpha}$. The Cayley–Laplace transform used in [1] is unitary from $L^2(\mathbb{R}_+)$ onto A_α^2 for every $\alpha > 0$, not only for $\alpha > 1$. One direct check uses the orthonormal Laguerre basis

$$e_n(x) = 2^{\alpha/2} \left(\frac{n!}{\Gamma(n + \alpha)} \right)^{1/2} x^{(\alpha-1)/2} e^{-x} L_n^{\alpha-1}(2x),$$

which is sent, up to harmless phases, to $c_\alpha(n)^{1/2} z^n$, the normalized monomial basis of A_α^2 . Write f for the disk image of ψ . The same algebra as on page 25 of [1] gives

$$|\langle \psi_{a,b}, \psi \rangle|^2 = V_f(z) := |f(z)|^2 (1 - |z|^2)^\alpha, \quad \frac{da db}{a^2} = 4\pi dm_h(z), \quad (1)$$

where z is the Cayley image of $ia - b$ and $dm_h = \pi^{-1}(1 - |z|^2)^{-2} dA(z)$. Normalized kernels

$$F_w(z) = \frac{(1 - |w|^2)^{\alpha/2}}{(1 - z\bar{w})^\alpha}$$

correspond exactly to affine coherent states. The RKHS point bound gives

$$0 \leq V_f(z) \leq \|f\|_{\alpha,2}^2. \quad (2)$$

Step 2: the renormalized norm constraint. Assume $\|f\|_{\alpha,2} = 1$ and set

$$U_f(z) = |f(z)|^{2/\alpha}(1 - |z|^2), \quad \mu(t) = m_h\{U_f > t\}, \quad G(t) = t(\mu(t) + 1).$$

Notice that $V_f = U_f^\alpha$. Lemma 3.1 of [2], originating in Kulikov's hyperbolic level-set theorem, says that G is nonincreasing on $(0, t_0)$, where $t_0 = \max U_f$. Thus $\nu = -dG$ is a positive Stieltjes measure and $G(t_0) = t_0$. By (2), $t_0 \leq 1$.

The decisive new input is Lemma 3.2 and the endpoint passage following the proof of Theorem 1.3 in [2]. Specializing their parameters to $\sigma = 2/\alpha$ gives the exact identity

$$1 = t_0^\alpha + \alpha \int_{(0,t_0)} t^{\alpha-1} d\nu(t). \quad (3)$$

For functions analytic past the circle this is precisely Lemma 3.2, proved by analytic continuation from the Bergman range. For general $f \in A_\alpha^2$ it follows by their dilation argument; equivalently, apply the identity to $f(\rho z)$ along radii avoiding boundary zeros and pass to the limit. This is the renormalized substitute for the divergent Bergman area norm.

Step 3: all hinge inequalities. Fix $0 < \tau < 1$ and define

$$H_f(\tau) = \int_{\mathbb{D}} (V_f - \tau^\alpha)_+ dm_h.$$

If $t_0 \leq \tau$, then $H_f(\tau) = 0$. Suppose $\tau < t_0$. Layer cake and $\mu(t) = G(t)/t - 1$ give

$$H_f(\tau) = \int_\tau^{t_0} \alpha t^{\alpha-2} (G(t) - t) dt. \quad (4)$$

Since $G(t) = t_0 + \nu([t, t_0])$, Tonelli rewrites this as a baseline term plus the ν -average of

$$K_\tau(y) = \frac{\alpha}{1-\alpha} (\tau^{\alpha-1} - y^{\alpha-1}), \quad y \geq \tau.$$

Relative to the norm cost $\alpha y^{\alpha-1}$, this kernel has ratio

$$R_\tau(y) = \frac{(\tau/y)^{\alpha-1} - 1}{1-\alpha}, \quad R'_\tau(y) = \tau^{\alpha-1} y^{-\alpha} > 0.$$

It is therefore bounded by $R_\tau(t_0)$. Using (3), including the harmless mass below τ in the cost, yields

$$H_f(\tau) \leq Q_\tau(t_0),$$

where elementary simplification gives

$$Q_\tau(T) = \tau^{\alpha-1} \left(\frac{T^{1-\alpha}}{1-\alpha} - T \right) - \frac{1}{1-\alpha} + \tau^\alpha. \quad (5)$$

The entire extremal problem is now the derivative

$$Q'_\tau(T) = \tau^{\alpha-1} (T^{-\alpha} - 1) \geq 0 \quad (0 < T \leq 1).$$

Consequently

$$\int_{\mathbb{D}} (V_f - s)_+ dm_h \leq \int_{\mathbb{D}} (V_{F_0} - s)_+ dm_h, \quad 0 < s < 1, \quad (6)$$

after setting $s = \tau^\alpha$; here $F_0 = 1$ and $V_{F_0}(z) = (1 - |z|^2)^\alpha$. Dilation and Fatou also give (6) directly if one begins with the past-the-boundary form of (3).

Step 4: convex majorization and the constant. If the coherent integral of a finite-valued convex Φ is finite, its constant and right-linear terms at zero must vanish. Hence, for a positive Borel measure λ on $(0, 1)$,

$$\Phi(v) = \int_{(0,1)} (v - s)_+ d\lambda(s), \quad 0 \leq v \leq 1.$$

Integrating (6) against λ proves

$$\int_{\mathbb{D}} \Phi(V_f) dm_h \leq \int_{\mathbb{D}} \Phi((1 - |z|^2)^\alpha) dm_h.$$

With $r = |z|$, then $t = 1 - r^2$ and $s = t^\alpha$, the right side is

$$\frac{1}{\alpha} \int_0^1 \Phi(s) s^{-1/\alpha-1} ds.$$

Multiplying by 4π in (1), and using $\alpha = 2\beta$, gives exactly the constant $2\pi/\beta$ in the theorem. Every normalized kernel has the same distribution, so every coherent state attains equality.

Step 5: equality. Corollaries 1.2 and 1.4 of [2] give the sharp point estimate and uniform boundary decay. Thus V_f attains its maximum; if that maximum is 1, equality in the RKHS point estimate forces f to be a normalized kernel. For every other unit vector, $t_0 < 1$. Equations (5)–(6) are then strict at every threshold $s \in (0, 1)$. A non-affine convex function has nonzero curvature measure in $(0, 1)$, so its integrated inequality is strict. This proves the optimizer classification. $\square$

Why this is literature-implied. The 2025 paper proves the Dirichlet-range norm identity and contractive power inequalities, but it neither cites arXiv:2210.14798 nor states the affine coherent-state convex majorization. The answer follows only after the disk identification and the hinge extremal calculation above. Accordingly this packet records a full implication of known literature, not an explicit later answer and not a priority claim for a new theorem.

Verification, search, and scope. The algebra in (5) and its derivative are checked by `code/verify_extremal_algebra.py`. Before the proof was found, Gamma-window formulas, an exact Hessian calculation, and polynomial searches for real powers and hinges found no competing extremizer. A bounded search through 2026-08-13 covered the exact quoted question, its affine/RKHS reformulation, local run indexes, and arXiv:1906.00223, 2112.09962, 2205.07998, 2211.03655, and 2510.14333. No explicit later full answer was found. The theorem addresses the pure-state Theorem 8 question. It does not claim the omitted density-matrix analogue, nor assign values to signed divergent integrals.

References

- [1] R. L. Frank, *Sharp inequalities for coherent states and their optimizers*, arXiv:2210.14798 (2022), especially Theorem 8 and Remark 20.
- [2] O. F. Brevig, A. Kulikov, K. Seip, and I. Zlotnikov, *Contractive Hardy–Littlewood inequalities in the Dirichlet range*, arXiv:2510.14333 (2025), especially Theorem 1.1, Corollaries 1.2 and 1.4, and Lemmas 3.1–3.2.
- [3] A. Kulikov, *Functionals with extrema at reproducing kernels*, Geom. Funct. Anal. **32** (2022), 938–949; arXiv:2203.12349.